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A Fast and Accurate Linear Constrained Zonotopic Set-membership Filter

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***Abstract*—This article proposes a fast and accurate linear constrained zonotopic set-membership filter (CZ-SMF), where the interval hulls of the estimates are computed online in real time. Unlike existing methods solving interval hulls independently and inefficiently at different time steps, we effectively leverage the temporal relations inherent in the recursive filtering process. More specifically, we establish two fundamental properties of linear CZ-SMFs: (i) structural invariance, which provides invariant parameters in constrained zonotopic estimates over time, and (ii) parametric similarity that guarantees the uniform boundedness of the time-varying optimization parameters for consecutive time steps. Inspired by the fundamental properties, we develop the structural-parametric information-based SMF (SPIN-SMF), which offers two important advantages. First, our designed optimization algorithm only involves linear operations by exploiting the structural invariance to precompute matrix inversions. Second, utilizing the parametric similarity, we propose a warm-start strategy to efficiently initialize the optimization process. Baseline and Monte Carlo simulations show that the SPIN-SMF reduces computation time by 98.72%-99.98% compared to the classical SMFs while maintaining high accuracy.**

***Index Terms*—Set-membership filter, constrained zonotope, fast and accurate algorithm.**

I. Introduction

Constrained zonotopic set-membership filter (CZ-SMF)[1] has emerged as a key framework for robust state estimation in dynamical systems with bounded noises and limited statistical properties [1]–[3]. With high estimation accuracy [4], the CZ-SMF has demonstrated its great potential in applications such as localization [5] and fault diagnosis [6].

Despite its accuracy advantages, the CZ-SMF faces significant computational bottlenecks in practice. Specifically, the real-time applicability of the CZ-SMF depends on solving the interval hulls for the constrained zonotopic estimates efficiently and accurately [4]; however, efficiently solving interval hulls is challenging, even for linear systems, as the complexity of constrained zonotopes grows unboundedly due to recursive prediction-update operations [1].

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[1]The CZ-SMF is an important class of set-membership filters, where the constrained zonotopes characterize the system uncertainty (including noises and states).

To deal with the growth of complexity while maintaining guaranteed estimates, current research has developed various outer-approximation order reduction techniques. These techniques approximate high-complexity constrained zonotopes with lower-complexity ones that tightly contain them. Geometrically, constrained zonotopes are described by generators (the same as traditional zonotopes [7]) and linear constraints. Based on the geometric properties of constrained zonotopes, [1] proposed order reduction methods for the constraints (associated with a rescaling strategy) and generators, respectively. Note that for the generator reduction, a lift-then-reduce strategy [1] can be employed, allowing the order reduction techniques for traditional zonotopes to be applied directly: for example, the first systematic method for generator reduction [7], seminal generator sorting strategies [8] and [9], and a principal component analysis (PCA)-based reduction method [10]. Different from the above geometry-based methods, an Observation-Information Tower-inspired method (OIT-CZ SMF) [11] was developed recently, by only utilizing the properties of dynamical systems to achieve efficient order reduction. These methods effectively address the complexity-growth issue and have been widely used [2], [3], [6].

While existing linear CZ-SMFs achieved a balance between estimation accuracy and computational efficiency for solving interval hulls at individual time steps, their relations caused by the recursive filtering process were totally and implicitly overlooked. This leads to low efficiency when sequentially solving interval hulls: 1) existing methods did not utilize (or even disrupt[2]) common/invariant structures across the estimated sets in different time steps, which could have significantly accelerated the computations; 2) solving interval hulls independently ignores the inherent parametric similarity between the optimization problems at consecutive time steps, where the similarity can help remove the redundancy in computations. Thus, it is necessary to seek fundamental properties of sequentially solving interval hulls for linear CZ-SMFs and develop fast and accurate algorithms accordingly.

In this work, we propose a fast and accurate CZ-SMF for linear time-invariant systems. The key novelty over existing methods (e.g., [1], [7]–[11]) is to exploit the temporal relation of recursive interval hull computations. It substantially reduces online computational complexity while preserving high accuracy. Our contributions are as follows:

- We establish two fundamental properties of linear CZ-

[2]The connections between the estimated sets can be cut off when the geometric descriptions of constrained zontopes are improperly simplified.

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SMFs. The first is structural invariance, as the prediction-update operations are temporally similar over time. It provides the common/invariant parameters in the constructed constrained zonotopic estimates at different time steps. The second is parametric similarity in sequentially solving interval hulls, given that the other parameters are invariant from the structural invariance. We rigorously prove the uniform boundedness of the time-varying parameters in consecutive time steps.

- Inspired by the established fundamental properties, we propose a structural-parametric information-based SMF (SPIN-SMF). First, by applying the structural invariance to precompute the matrix inversions, a fast coordinate descent augmented Lagrangian algorithm is developed, which only involves linear operations. Second, motivated by the parametric similarity, we design a warm-start strategy to efficiently reuse the information of the previous time step. It constructs a guaranteed initial point to significantly accelerate the convergence of the algorithm. Simulations show that the SPIN-SMF reduces computation time by $98.72\% - 99.98\%$ compared to the classical SMFs while maintaining high accuracy.

Notation: Throughout this article, $\mathbb{R}$, $\mathbb{N}_0$, and $\mathbb{N}$ denote the sets of real numbers, nonnegative integers, and positive integers, respectively. $\mathbb{R}^n$ stands for the n-dimensional Euclidean space. For $a, b \in \mathbb{R}^n$, $a \leq b$ means that each component of a is not greater than that of b. For an uncertain variable $\mathbf{x}$ defined on a sample space Ω, its range is $[\![\mathbf{x}]\!] = \{\mathbf{x}(\omega)\colon \omega \in \Omega\}$ and its realization is $x = \mathbf{x}(\omega)$ [12], [13]. We use $\|\cdot\|_\infty$ to represent the infinity norm (of a vector) or the row sum norm (of a matrix). The set $\mathcal{S}_k$ is *uniformly bounded* (w.r.t. $k \in \mathcal{K}$) if there exists $d_s > 0$ such that $d(\mathcal{S}_k) \leq d_s$ for all $k \in \mathcal{K}$, where $d(\mathcal{S}_k) = \sup_{s,s' \in \mathcal{S}_k} \|s - s'\|_\infty$ is the diameter of $\mathcal{S}_k$. The interval hull of a bounded set $\mathcal{S} \ni s = (s^{(1)}, \ldots, s^{(n)})$ is $\overline{\mathrm{IH}}(\mathcal{S}) = \prod_{i=1}^n [\underline{s}^{(i)}, \overline{s}^{(i)}]$, where $\underline{s}^{(i)} = \inf_{s^{(i)}}\{s \in \mathcal{S}\}$ and $\overline{s}^{(i)} = \sup_{s^{(i)}}\{s \in \mathcal{S}\}$. Given a matrix $M \in \mathbb{R}^{m \times n}$, the Moore-Penrose inverse and the kernel (null space) are M^+ and $\ker(M)$, respectively; $\rho(\cdot)$ returns the spectral radius of a square matrix; I_n is an identity matrix of size n and $[\,]$ is a null matrix. The operator $\circ$ is the composition of two maps.

II. Problem Formulation

Consider the following discrete-time linear system described by uncertain variables [12] ($k \in \mathbb{N}_0$):

$$\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{w}_k, \tag{1}$$

$$\mathbf{y}_k = C\mathbf{x}_k + \mathbf{v}_k, \tag{2}$$

where (1) and (2) are the state and measurement equations, respectively. The state equation describes how the system state $x_k \in [\![\mathbf{x}_k]\!] \subseteq \mathbb{R}^n$ changes over time, affected by the process noise $w_k \in [\![\mathbf{w}_k]\!] \subseteq \mathbb{R}^p$. The measurement equation gives how the system state is measured from the measurement $y_k \in [\![\mathbf{y}_k]\!] \subseteq \mathbb{R}^m$, and $v_k \in [\![\mathbf{v}_k]\!] \subseteq \mathbb{R}^m$ represents the measurement noise; $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times p}$, and $C \in \mathbb{R}^{m \times n}$.

We assume that $\forall k \in \mathbb{N}_0$, the process noises $\mathbf{w}_{0:k}$, the measurement noises $\mathbf{v}_{0:k}$, and the initial state $\mathbf{x}_0$ are mutually unrelated[3]. This makes the system satisfy a non-stochastic hidden Markov model, which guarantees the optimality of the classical SMFing [13], provided in Lemma 1.

Lemma 1 (Optimal Linear SMFing)**.** *For the system described by* (1) *and* (2)*, the optimal estimate*[4] $[\![\mathbf{x}_k|y_{0:k}]\!]$ *can be recursively derived by the following steps.*

- *Initialization ($k = 0$): Set the initial prior set* $[\![\mathbf{x}_0]\!]$.
- *Prediction ($k \in \mathbb{N}$): Given the estimate (posterior set)* $[\![\mathbf{x}_{k-1}|y_{0:k-1}]\!]$ *derived at $k-1$, the prior set at k is*

$$[\![\mathbf{x}_k|y_{0:k-1}]\!] = A[\![\mathbf{x}_{k-1}|y_{0:k-1}]\!] \oplus B[\![\mathbf{w}_{k-1}]\!], \tag{3}$$

  where $\oplus$ denotes the Minkowski sum.
- *Update ($k \in \mathbb{N}_0$): Given the measurement y_k and the prior set obtained in* (3)*, the posterior set* $[\![\mathbf{x}_k|y_{0:k}]\!]$ *is*

$$[\![\mathbf{x}_k|y_{0:k}]\!] = \mathcal{X}_k(C, y_k, [\![\mathbf{v}_k]\!]) \bigcap [\![\mathbf{x}_k|y_{0:k-1}]\!], \tag{4}$$

  where $[\![\mathbf{x}_0|y_{0:-1}]\!] := [\![\mathbf{x}_0]\!]$ *and*

$$\mathcal{X}_k(C, y_k, [\![\mathbf{v}_k]\!]) = \ker(C) \oplus C^+(\{y_k\} \oplus [\![-\mathbf{v}_k]\!]). \tag{5}$$

In this work, we consider SMFs with constrained zonotopic descriptions [1], where $[\![\mathbf{x}_k|y_{0:k}]\!]$ are represented by constrained zonotopes to perform prediction/update set operations[5]. For computational purposes, an extended version of the constrained zonotope is given in Definition 1.

Definition 1 (Extended Constrained Zonotope [11])**.** *A set $\mathcal{Z} \subseteq \mathbb{R}^n$ is an extended constrained zonotope if there exists a quintuple $(\hat{G}, \hat{c}, \hat{A}, \hat{b}, \hat{h}) \in \mathbb{R}^{n \times n_g} \times \mathbb{R}^n \times \mathbb{R}^{n_c \times n_g} \times \mathbb{R}^{n_c} \times \mathbb{R}^{n_g}_+$ such that $\mathcal{Z}$ is expressed by*

$$Z(\hat{G}, \hat{c}, \hat{A}, \hat{b}, \hat{h}) := \left\{\hat{G}\xi + \hat{c}\colon \hat{A}\xi = \hat{b},\ -\hat{h} \leq \xi \leq \hat{h}\right\},$$

where n_g and n_c are the generator and constraint dimensions [1], respectively.

The extended constrained zonotope and the (standard) constrained zonotope in [1] are mathematically equivalent. However, the former has better numerical stability, as $\hat{h}$ helps maintain a good condition number for $\hat{A}$.

With a slight abuse of notation, we define an extended zonotope by $Z(\hat{G}, \hat{c}, \hat{h}) := \left\{\hat{G}\xi + \hat{c}\colon\ -\hat{h} \leq \xi \leq \hat{h}\right\}$. For a unified algebraic representation, we describe the interval hull of an extended constrained zonotope $\mathcal{Z} \in \mathbb{R}^n$ by using the above extended zonotopic form:

$$\overline{\mathrm{IH}}(\mathcal{Z}) = [\underline{\rho}, \overline{\rho}] = Z(I_n, \square\hat{c}, \square\hat{h}), \tag{6}$$

where $\square\hat{c} = (\underline{\rho} + \overline{\rho})/2$ and $\square\hat{h} = (\overline{\rho} - \underline{\rho})/2$.

Under Definition 1, the initial set $[\![\mathbf{x}_0]\!]$, process noise sets $[\![\mathbf{w}_k]\!]$, and measurement noise sets $[\![\mathbf{v}_k]\!]$ are constructed by extended constrained zonotopes as follows

$$x_0 \in [\![\mathbf{x}_0]\!] = Z(\hat{G}_0^-, \hat{c}_0^-, \hat{A}_0^-, \hat{b}_0^-, \hat{h}_0^-), \tag{7a}$$

$$w_k \in [\![\mathbf{w}_k]\!] = Z(\hat{G}_\mathbf{w}, \hat{c}_\mathbf{w}, \hat{A}_\mathbf{w}, \hat{b}_\mathbf{w}, \hat{h}_\mathbf{w}), \tag{7b}$$

$$v_k \in [\![\mathbf{v}_k]\!] = Z(\hat{G}_\mathbf{v}, \hat{c}_\mathbf{v}, \hat{A}_\mathbf{v}, \hat{b}_\mathbf{v}, \hat{h}_\mathbf{v}), \tag{7c}$$

[3]Uncertain variables $\mathbf{x}_1, \ldots, \mathbf{x}_r$ are mutually unrelated if $[\![\mathbf{x}_1, \ldots, \mathbf{x}_r]\!] = [\![\mathbf{x}_1]\!] \times \cdots \times [\![\mathbf{x}_r]\!]$, where $\times$ is the Cartesian product.

[4]The optimal estimate is the smallest set containing all possible states [13].

[5]Here $[\![\mathbf{x}_k|y_{0:k}]\!]$ can be viewed as a bounded support set (the set of admissible realizations), which aligns with the support-propagation interpretation of zonotopic filtering discussed in [14].

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where $[\![\mathbf{x}_0]\!]$, $[\![\mathbf{w}_k]\!]$, and $[\![\mathbf{v}_k]\!]$ are bounded; we define $d_w = d([\![\mathbf{w}_k]\!])$ and $d_v = d([\![\mathbf{v}_k]\!])$. With (7), the prior set $[\![\mathbf{x}_k|y_{0:k-1}]\!]$ and the posterior set $[\![\mathbf{x}_k|y_{0:k}]\!]$ in Lemma 1 inherently preserve the constrained zonotopic structure [1] without approximations; thus, we call such an SMFing in Lemma 1 with exact constrained zonotopic descriptions the optimal linear CZ-SMF (constrained zonotopic set-membership filter). We write the optimal estimate set as

$$[\![\mathbf{x}_k|y_{0:k}]\!] = Z(\hat{G}_k^*, \hat{c}_k^*, \hat{A}_k^*, \hat{b}_k^*, \hat{h}_k^*). \tag{8}$$

However, the resulting extended constrained zonotope (8) cannot explicitly describe the interval of each state component [fully characterized by the interval hull (6)], and solving $2n$ linear programmings (LPs) is required [1], [4]; see Problem 1.

Problem 1 (Calculate the Exact Interval Hull)**.** *Calculate the interval hull of $[\![\mathbf{x}_k|y_{0:k}]\!]$ returned by the CZ-SMF for $k \in \mathbb{N}_0$:*

$$\overline{\mathrm{IH}}([\![\mathbf{x}_k|y_{0:k}]\!]) = [\underline{\rho}_k^*, \overline{\rho}_k^*] \overset{(6)}{=} Z(I_n, \square\hat{c}_k^*, \square\hat{h}_k^*), \tag{9}$$

where $\underline{\rho}_k^* = [\underline{\rho}_k^{*,1}, \ldots, \underline{\rho}_k^{*,n}]^{\mathrm{T}}$ and $\overline{\rho}_k^* = [\overline{\rho}_k^{*,1}, \ldots, \overline{\rho}_k^{*,n}]^{\mathrm{T}}$ are solved by LPs for each dimension $i = 1, \ldots, n$ as follows:

$$\mathcal{P}_k^*\colon \begin{matrix} \underline{\rho}_k^{*,i} = \min\limits_{\xi} \left(\hat{G}_k^{*,(i)}\xi + \hat{c}_k^{*,(i)}\right) \\ \overline{\rho}_k^{*,i} = \max\limits_{\xi} \left(\hat{G}_k^{*,(i)}\xi + \hat{c}_k^{*,(i)}\right) \end{matrix} \text{ s.t. } \begin{cases} \hat{A}_k^*\xi = \hat{b}_k^*, \\ -\hat{h}_k^* \le \xi \le \hat{h}_k^*, \end{cases}$$

where $\hat{G}_k^{*,(i)}$ is row i of $\hat{G}_k^*$ and $\hat{c}_k^{*,(i)}$ is component i of $\hat{c}_k^*$.

Note that the CZ-SMF suffers from unbounded increases of the generator and constraint dimensions (i.e., n_g and n_c in Definition 1) due to successive extended constrained zonotope operations [1]. This fact makes solving Problem 1 computationally intractable as time elapses. Although existing order reduction techniques (e.g., [1], [7]–[11], [15]–[18]) can balance the efficiency and accuracy by reducing n_g and n_c for each $\mathcal{P}_k^*$, the relationship among $\mathcal{P}_k^*$ for different k is overlooked, which results in:

- **Invariant Structure Breaking:** Even though $\mathcal{P}_k^*$ varies over time, these problems share common/invariant structures (analyzed in Section III-A), which can inspire offline calculated parts to significantly enhance efficiency (designed in Section IV-A). However, the existing methods calculate these problems independently, which does not utilize the invariant structure or even breaks it.
- **Redundancy in Sequential Computation:** Apart from the invariant structure, existing methods ignore the parametric similarity between consecutive optimization problems in recursive filtering processes (analyzed in Section III-B), which leads to redundant computations. Indeed, when solving an optimization for a new time step, the solution from the previous time step can be leveraged due to their similarity (see the warm-start strategy designed in Section IV-B).

In this article, we focus on efficiently and accurately solving Problem 1 at each $k \in \mathbb{N}_0$. First, we provide two fundamental properties (i.e., the structural invariance and parametric similarity) in Section III. Based on these two properties, a fast and accurate algorithm is designed in Section IV.

III. Fundamental Properties

This section establishes two fundamental properties: structural invariance of the linear CZ-SMF process and parametric similarity between sequential optimization problems.

A. Structural Invariance

To facilitate analysis, we introduce the recursion operator in Definition 2.

Definition 2 (Recursion Operator)**.** *For $i \in \mathbb{N}_0$, the prediction-update operator determined by* (3) *and* (4) *is ($\mathcal{S} \subseteq \mathbb{R}^n$)*

$$g_i : \begin{cases} \mathcal{S} \mapsto \mathcal{X}_i(C, y_i, [\![\mathbf{v}_i]\!]) \bigcap (A\mathcal{S} \oplus B[\![\mathbf{w}_{i-1}]\!]), & i > 0, \\ \mathcal{S} \mapsto \mathcal{X}_i(C, y_i, [\![\mathbf{v}_i]\!]) \bigcap \mathcal{S}, & i = 0. \end{cases}$$

For $k \in \mathbb{N}$, a recursion operator $R_{k,i}$, describing any set $\mathcal{S}$ at i under recursive prediction-update operations up to k, is

$$R_{k,i}(\mathcal{S}) = g_k \circ \cdots \circ g_{i+1}(\mathcal{S}),\ k > i. \tag{10}$$

Based on Definition 2, the optimal linear CZ-SMF is

$$[\![\mathbf{x}_k|y_{0:k}]\!] = R_{k,k-\delta}([\![\mathbf{x}_{k-\delta}|y_{0:k-\delta}]\!]), \quad k \ge \delta, \tag{11}$$

where $\delta \in \mathbb{N}_0$ and

$$\begin{aligned} [\![\mathbf{x}_{k-\delta}|y_{0:k-\delta}]\!] &= R_{k-\delta,0}([\![\mathbf{x}_0|y_0]\!]) \\ &= Z(\hat{G}_{k-\delta}^*, \hat{c}_{k-\delta}^*, \hat{A}_{k-\delta}^*, \hat{b}_{k-\delta}^*, \hat{h}_{k-\delta}^*). \end{aligned} \tag{12}$$

Note that (11) divides the filtering process into two parts:

(i) the historical posterior $[\![\mathbf{x}_{k-\delta}|y_{0:k-\delta}]\!]$ is derived at $k-\delta$;
(ii) the recursion operator $R_{k,k-\delta}$ describes the prediction-update operations over the time window $\{k-\delta+1, \ldots, k\}$ with length δ.

With these two parts, we give the structural invariance of the optimal linear CZ-SMF in (11); see Proposition 1.

Proposition 1 (Structural Invariance)**.** *$\forall k, \delta \in \mathbb{N}_0$ with $k \ge \delta$, the optimal estimate $[\![\mathbf{x}_k|y_{0:k}]\!]$ [see* (8)*] has the parameters $(\hat{G}_k^*, \hat{c}_k^*, \hat{A}_k^*, \hat{b}_k^*, \hat{h}_k^*) = (\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k)$ in* (13) *as follows*

$$\hat{G}_k = \begin{bmatrix} A^\delta\phi_G & A^\delta B\hat{G}_{\mathbf{w}} & 0 & A^{\delta-1}B\hat{G}_{\mathbf{w}} & \ldots & B\hat{G}_{\mathbf{w}} & 0 \end{bmatrix},$$

$$\hat{c}_k = A^\delta\phi_c + \sum_{r=k-\delta}^{k-1} A^{k-1-r}B\hat{c}_{\mathbf{w}},$$

$$\hat{A}_k = \begin{bmatrix} \phi_A & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \hat{A}_{\mathbf{w}} & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \hat{A}_{\mathbf{v}} & 0 & 0 & \cdots & 0 & 0 \\ CA\phi_G & CB\hat{G}_{\mathbf{w}} & \hat{G}_{\mathbf{v}} & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \hat{A}_{\mathbf{w}} & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & \hat{A}_{\mathbf{v}} & \cdots & 0 & 0 \\ CA^2\phi_G & CAB\hat{G}_{\mathbf{w}} & 0 & CB\hat{G}_{\mathbf{w}} & \hat{G}_{\mathbf{v}} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & \hat{A}_{\mathbf{w}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & \hat{A}_{\mathbf{v}} \\ CA^\delta\phi_G & CA^{\delta-1}B\hat{G}_{\mathbf{w}} & 0 & 0 & 0 & \cdots & CB\hat{G}_{\mathbf{w}} & \hat{G}_{\mathbf{v}} \end{bmatrix},$$

$$\hat{b}_k = \begin{bmatrix} \phi_b & \hat{b}_{\mathbf{w}} & \hat{b}_{\mathbf{v}} & \hat{b}_{y_{k-\delta}} & \ldots & \hat{b}_{\mathbf{w}} & \hat{b}_{\mathbf{v}} & \hat{b}_{y_l} & \ldots & \hat{b}_{y_k} \end{bmatrix}^{\mathrm{T}},$$

$$\hat{h}_k = \begin{bmatrix} \phi_h & \hat{h}_{\mathbf{w}} & \hat{h}_{\mathbf{v}} & \ldots & \hat{h}_{\mathbf{w}} & \hat{h}_{\mathbf{v}} & \ldots & \hat{h}_{\mathbf{w}} & \hat{h}_{\mathbf{v}} \end{bmatrix}^{\mathrm{T}},$$

$$\hat{b}_{y_l} = y_l - \hat{c}_{\mathbf{v}} - C(A^{l-k+\delta}\phi_c + \sum_{r=k-\delta}^{l-1} A^{l-1-r}B\hat{c}_{\mathbf{w}}), \tag{13}$$

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where $k-\delta \le l \le k$, $\hat{G}_k \in \mathbb{R}^{n\times n_g^k}$, $\hat{c}_k \in \mathbb{R}^n$, $\hat{A}_k \in \mathbb{R}^{n_c^k \times n_g^k}$, $\hat{b}_k \in \mathbb{R}^{n_c^k}$, $\hat{h}_k \in \mathbb{R}^{n_g^k}$, $n_g^k = n+\delta(p+m)$, *and* $n_c^k = \delta(p+2m)$. *The structure of the parameters is invariant with* k, *and the only time-varying parts are as follows.*

- *Historical posterior:* $\phi_G = \hat{G}^*_{k-\delta}$, $\phi_c = \hat{c}^*_{k-\delta}$, $\phi_A = \hat{A}^*_{k-\delta}$, $\phi_b = \hat{b}^*_{k-\delta}$, *and* $\phi_h = \hat{h}^*_{k-\delta}$.
- *Recursion operator:* $\{y_{k-\delta+1}, \dots, y_k\}$.

Proof: Equation (13) follows from (11), (12), and basic set operations of extended constrained zonotopes [1], [11]. The time-varying parts can be observed from (13). ∎

Due to the unbounded increases of n_g and n_c, calculating the exact $\overline{\mathrm{IH}}(\llbracket \mathbf{x}_k | y_{0:k} \rrbracket)$ in Problem 1 is computationally intractable for large k. When $k \ge \delta$, we can replace $\llbracket \mathbf{x}_{k-\delta} | y_{0:k-\delta} \rrbracket$ in (11) with a low-complexity interval hull to compute an outer bound $\mathcal{Z}_k = Z(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k) \supseteq \llbracket \mathbf{x}_k | y_{0:k} \rrbracket$ as follows

$$\mathcal{Z}_k = \begin{cases} R_{k,k-\delta}(\overline{\mathrm{IH}}(\mathcal{Z}_{k-\delta})), & k \ge \delta, \\ \llbracket \mathbf{x}_k | y_{0:k} \rrbracket, & 0 \le k < \delta, \end{cases} \tag{14}$$

where we set $\mathcal{Z}_k = \llbracket \mathbf{x}_k | y_{0:k} \rrbracket$ ($k < \delta$) for compactness, which implies $\hat{G}_k := \hat{G}^*_k$, $\hat{c}_k := \hat{c}^*_k$, $\hat{A}_k := \hat{A}^*_k$, $\hat{b}_k := \hat{b}^*_k$, and $\hat{h}_k := \hat{h}^*_k$ for $k < \delta$.

Note that the structural invariance in Proposition 1 also holds for (14). This motivates us to replace $\llbracket \mathbf{x}_k | y_{0:k} \rrbracket$ in Problem 1 with $\mathcal{Z}_k$ in (14), transforming[6] it into Problem 2.

Problem 2 (Calculate the Near Optimal Interval Hull)**.** *For* $k \in \mathbb{N}_0$, *calculate the interval hull of* $\mathcal{Z}_k$ *defined by* (14)*:*

$$\overline{\mathrm{IH}}(\mathcal{Z}_k) = [\underline{\rho}_k, \overline{\rho}_k] \overset{(6)}{=} Z(I_n, \square\hat{c}_k, \square\hat{h}_k), \tag{15}$$

where $\underline{\rho}_k = [\underline{\rho}^1_k, \dots, \underline{\rho}^n_k]^{\mathrm{T}}$ *and* $\overline{\rho}_k = [\overline{\rho}^1_k, \dots, \overline{\rho}^n_k]^{\mathrm{T}}$ *are solved by LPs for each dimension* $i = 1, \dots, n$ *as follows*[7]*:*

$$\mathcal{P}_k\colon \begin{array}{l} \underline{\rho}^i_k = \min\limits_{\eta,\zeta} \left(\hat{G}^{(i)}_k \eta + \hat{c}^{(i)}_k\right) \\ \overline{\rho}^i_k = \max\limits_{\eta,\zeta} \left(\hat{G}^{(i)}_k \eta + \hat{c}^{(i)}_k\right) \end{array} \text{ s.t. } \begin{cases} \hat{A}_k \eta = \hat{b}_k, \\ \eta = \zeta, \\ -\hat{h}_k \le \zeta \le \hat{h}_k, \end{cases}$$

where $\hat{G}^{(i)}_k$ *is row* i *of* $\hat{G}_k$ *and* $\hat{c}^{(i)}_k$ *is component* i *of* $\hat{c}_k$.

Transforming Problem 1 into Problem 2 has two advantages:

- **Utilizing Invariant Structure:** For $k \ge \delta$, by structural invariance, the parameters of the extended constrained zonotope $\mathcal{Z}_k = Z(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k)$ are in (13), where

$$\phi_G = I_n,\ \phi_c = \square\hat{c}_{k-\delta},\ \phi_A = [\ ],\ \phi_b = [\ ],\ \phi_h = \square\hat{h}_{k-\delta}. \tag{16}$$

  Since $\phi_G = I_n$ and $\phi_A = [\ ]$, the parameters $\hat{G}_k$ and $\hat{A}_k$ are time-invariant. This also fixes the generator and constraint dimensions of $\mathcal{Z}_k$ to n_g^k and n_c^k, respectively.
- **Utilizing Historical Solutions:** For $k \ge \delta$, the historical posterior $\overline{\mathrm{IH}}(\mathcal{Z}_{k-\delta})$ in (14) has already been derived at $k-\delta$. More specifically, for $\mathcal{P}_k$, the parameters $\square\hat{c}_{k-\delta}$ and $\hat{h}_{k-\delta}$ are directly from (15) at $k-\delta$, which does not rely on additional order reduction methods.

Next, we show that the computation of (14) is reasonable, i.e., $\mathcal{Z}_k$ remains uniformly bounded under the detectable pair (A, C). The result is established in Theorem 1, which relies on the observability decomposition of (1) and (2). The transformed (decomposed) system has the form

$$\begin{aligned} \begin{bmatrix} \tilde{\mathbf{x}}^o_{k+1} \\ \tilde{\mathbf{x}}^{\bar{o}}_{k+1} \end{bmatrix} &= \begin{bmatrix} \tilde{A}_o & 0 \\ \tilde{A}_{21} & \tilde{A}_{\bar{o}} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}^o_k \\ \tilde{\mathbf{x}}^{\bar{o}}_k \end{bmatrix} + \begin{bmatrix} \tilde{B}_o \\ \tilde{B}_{\bar{o}} \end{bmatrix} \mathbf{w}_k, \\ \mathbf{y}_k &= \tilde{C}_o \tilde{\mathbf{x}}^o_k + \mathbf{v}_k, \end{aligned} \tag{17}$$

where $\tilde{\mathbf{x}}^o_k = T_o \mathbf{x}_k$ and $\tilde{\mathbf{x}}^{\bar{o}}_k = T_{\bar{o}} \mathbf{x}_k$ are the states of the observable and unobservable subsystems, respectively. Here, $T = [T_o^\top \ T_{\bar{o}}^\top]^\top$ is the transformation matrix with $T^{-1} = \begin{bmatrix} U_o & U_{\bar{o}} \end{bmatrix}$ such that $\tilde{A}_o = T_o A U_o$, $\tilde{A}_{\bar{o}} = T_{\bar{o}} A U_{\bar{o}}$, $\tilde{A}_{21} = T_{\bar{o}} A U_o$, $\tilde{B}_o = T_o B$, $\tilde{B}_{\bar{o}} = T_{\bar{o}} B$, and $\tilde{C}_o = C U_o$.

Theorem 1 (Uniform Boundedness)**.** *For detectable* (A, C), *there exists a* $\underline{\delta} \in \mathbb{N}_0$ *satisfying*

$$\underline{\delta} > -\log_\gamma(M_\gamma \|T_{\bar{o}}\|_\infty \|T^{-1}\|_\infty) \tag{18}$$

such that $\forall \delta \ge \underline{\delta}$, $\mathcal{Z}_k$ *in* (14) *is uniformly bounded, the scalar* γ *satisfies* $\rho(\tilde{A}_{\bar{o}}) \le \gamma \le 1$ *and* $M_\gamma \ge 1$.

Proof: Let $\mathcal{Z}^o_k = T_o \mathcal{Z}_k$ and $\mathcal{Z}^{\bar{o}}_k = T_{\bar{o}} \mathcal{Z}_k$. We prove the uniform boundedness of $\mathcal{Z}_k$, i.e., $\exists \bar{d} \ge 0$ such that $d(\mathcal{Z}_k) \le \|T^{-1}\|_\infty \max\{d(\mathcal{Z}^o_k), d(\mathcal{Z}^{\bar{o}}_k)\} \le \bar{d}$ for $k \ge \delta$. By Proposition 4 in [11], $\exists \bar{d}_o \ge 0$ such that $d(\mathcal{Z}^o_k) \le \bar{d}_o$. Thus, we only focus on $d(\mathcal{Z}^{\bar{o}}_k)$. With (14) and (17), we have

$$\begin{aligned} d(\mathcal{Z}^{\bar{o}}_k) \le{}& \|\tilde{A}^\delta_{\bar{o}}\|_\infty \|T_{\bar{o}}\|_\infty d(\overline{\mathrm{IH}}(\mathcal{Z}_{k-\delta})) + \\ & \sum_{j=k-\delta}^{k-1} \|\tilde{A}^{k-j-1}_{\bar{o}}\|_\infty (\|\tilde{A}_{21}\|_\infty d(\mathcal{Z}^o_j) + \|\tilde{B}_{\bar{o}}\|_\infty d_w). \end{aligned} \tag{19}$$

Since (A, C) is detectable, the unobservable part $\tilde{A}_{\bar{o}}$ is Schur stable and hence $\rho(\tilde{A}_{\bar{o}}) < 1$. By Lemma 3 in [19], $\forall \gamma \in]\rho(\tilde{A}_{\bar{o}}), 1[$, $\exists M_\gamma \ge 1$ such that $\|\tilde{A}^\delta_{\bar{o}}\|_\infty \le M_\gamma \gamma^\delta$ holds for $\delta \in \mathbb{N}_0$. Since $d(\mathcal{Z}^o_j) \le \bar{d}_o$, $\sum_{j=k-\delta}^{k-1} \gamma^{k-j-1} \le \frac{1}{1-\gamma}$, and

$$d(\overline{\mathrm{IH}}(\mathcal{Z}_{k-\delta})) = d(\mathcal{Z}_{k-\delta}) \le \|T^{-1}\|_\infty \max\{\bar{d}_o, d(\mathcal{Z}^{\bar{o}}_{k-\delta})\},$$

the RHS of (19) can be upper bounded by

$$d(\mathcal{Z}^{\bar{o}}_k) \le M_\gamma \gamma^\delta \|T_{\bar{o}}\|_\infty \|T^{-1}\|_\infty \max\{\bar{d}_o, d(\mathcal{Z}^{\bar{o}}_{k-\delta})\} + \Theta,$$

where $\Theta = M_\gamma(\|\tilde{A}_{21}\|_\infty \bar{d}_o + \|\tilde{B}_{\bar{o}}\|_\infty d_w)/(1-\gamma)$. Let $K := M_\gamma \|T_{\bar{o}}\|_\infty \|T^{-1}\|_\infty$. Since $0 < \gamma < 1$, $\lim_{\delta\to\infty} K\gamma^\delta = 0$. Hence, $\exists \underline{\delta} \in \mathbb{N}_0$ such that $K\gamma^{\underline{\delta}} = M_\gamma \gamma^{\underline{\delta}} \|T_{\bar{o}}\|_\infty \|T^{-1}\|_\infty < 1$. Moreover, for all $\delta \ge \underline{\delta}$, $\gamma^\delta \le \gamma^{\underline{\delta}}$, and thus $M_\gamma \gamma^\delta \|T_{\bar{o}}\|_\infty \|T^{-1}\|_\infty \le M_\gamma \gamma^{\underline{\delta}} \|T_{\bar{o}}\|_\infty \|T^{-1}\|_\infty < 1$. This implies $\limsup_{k\to\infty} d(\mathcal{Z}^{\bar{o}}_k) < \infty$ (since $\max\{\bar{d}_o, d(\mathcal{Z}^{\bar{o}}_{k-\delta})\} \le \bar{d}_o + d(\mathcal{Z}^{\bar{o}}_{k-\delta})$), and thus $\mathcal{Z}_k$ is uniformly bounded. ∎

As a complement to Theorem 1, our simulation shows that the estimated accuracy loss when transforming Problem 1 into Problem 2 is very small (see Section IV-B). Structural invariance inspires us to design a fast algorithm for Problem 2 (see Section IV-A), i.e., common/invariant parts are computed offline so that the online computation involves only linear operations.

[6]This transformation typically incurs only a small accuracy loss; see Section V for the accuracy gap between the SPIN SMF and the optimal SMF. In the future, we will quantify the accuracy loss incurred when transforming Problem 1 into Problem 2.

[7]Introducing $\eta = \zeta$ can effectively handle the box constraint $-\hat{h}_k \le \zeta \le \hat{h}_k$; see Section IV.

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B. Parametric Similarity

In Section III-A, we know that for different k ($k \geq \delta$), the only time-varying parameters affecting the solution to $\mathcal{P}_k$ are $\hat{b}_k$ and $\hat{h}_k$.[8] In this subsection, we show that the parametric difference between $\mathcal{P}_k$ and $\mathcal{P}_{k+1}$ is uniformly bounded; see Theorem 2. As a result, the solution to $\mathcal{P}_k$ can provide useful information when solving $\mathcal{P}_{k+1}$, which inspires a warm-start strategy in Section IV-B to further improve the efficiency.

Theorem 2 (Parametric Similarity)**.** *Let* $\Delta\hat{b}_k = \hat{b}_{k+1} - \hat{b}_k$, $\Delta\hat{h}_k = \hat{h}_{k+1} - \hat{h}_k$, $\|\mathbf{w}_k\|_\infty \leq d_w$, $\|\mathbf{v}_k\|_\infty \leq d_v$, *and* $\bar{d} = \sup_{k\geq\delta} d(\mathcal{Z}_k)$. $\forall k \geq \delta$, *we have*

$$\|\Delta\hat{b}_k\|_\infty \leq \max_{k-\delta\leq l\leq k}\{\|CA^{l-k+\delta}\|_\infty\}\bar{d} + \Psi,\ \|\Delta\hat{h}_k\|_\infty \leq \frac{\bar{d}}{2}, \quad (20)$$

where $\Psi = 2(\sum_{j=0}^{\delta-1}\|CA^jB\|_\infty d_w + d_v)$ *is the noise-induced bound constant.*

Proof: Based on (13), we can derive

$$\Delta\hat{b}_k = \begin{bmatrix}0 & 0 & 0 & \Delta\hat{b}^k_{y_{k-\delta}} & \cdots & \Delta\hat{b}^k_{y_l} & \cdots & \Delta\hat{b}^k_{y_k}\end{bmatrix}^{\mathrm{T}}, \quad (21)$$

where $\Delta\hat{b}^k_{y_l} = \hat{b}^{k+1}_{y_{l+1}} - \hat{b}^k_{y_l}$; thus, we have

$$\begin{aligned}\|\Delta\hat{b}^k_{y_l}\|_\infty &= \|y_{l+1} - y_l - CA^{l-k+\delta}(\square\hat{c}_{k+1-\delta} - \square\hat{c}_{k-\delta})\|_\infty\\ &\overset{(2)}{=} \|C(x_{l+1} - x_l) - \Delta v_l - CA^{l-k+\delta}(\square\hat{c}_{k+1-\delta} - \square\hat{c}_{k-\delta})\|_\infty\\ &\overset{(1)}{=} \|CA^{l-k+\delta}[(x_{k+1-\delta} - \square\hat{c}_{k+1-\delta}) - (x_{k-\delta} - \square\hat{c}_{k-\delta})]\\ &\quad + C\sum_{j=k-\delta}^{l-1} A^{l-1-j}B(w_{j+1} - w_j) + \Delta v_l\|_\infty,\end{aligned}$$

where $\Delta v_l = (v_{l+1} - v_l)$. Since $d(\mathcal{Z}_k) = d(\overline{\mathrm{IH}}(\mathcal{Z}_k)) = \sup_{z,z'\in\overline{\mathrm{IH}}(\mathcal{Z}_k)}\|z - z'\|_\infty$ and $x_k, \square\hat{c}_k \in \overline{\mathrm{IH}}(\mathcal{Z}_k)$, we get $\|\Delta\hat{b}^k_{y_l}\|_\infty \leq \|CA^{l-k+\delta}\|_\infty\bar{d} + 2\sum_{j=k-\delta}^{l-1}\|CA^{l-1-j}B\|_\infty d_w + 2d_v$. Let $\Psi = 2(\sum_{j=0}^{\delta-1}\|CA^jB\|_\infty d_w + d_v)$. Then, $\|\Delta\hat{b}_k\|_\infty = \max_{k-\delta\leq l\leq k}\{\|\Delta\hat{b}^k_{y_l}\|_\infty\}$ and $\max_{k-\delta\leq l\leq k}\{\|\Delta\hat{b}^k_{y_l}\|_\infty\} \leq \max_{k-\delta\leq l\leq k}\{\|CA^{l-k+\delta}\|_\infty\}\bar{d} + \Psi$.

Similarly, based on (13), $\Delta\hat{h}_k$ is given by

$$\Delta\hat{h}_k = \begin{bmatrix}(\square\hat{h}_{k+1-\delta} - \square\hat{h}_{k-\delta}) & 0 & \cdots & 0\end{bmatrix}^{\mathrm{T}}, \quad (22)$$

and $\|\Delta\hat{h}_k\|_\infty$ can be upper bounded by

$$\|\Delta\hat{h}_k\|_\infty = \|\square\hat{h}_{k+1-\delta} - \square\hat{h}_{k-\delta}\|_\infty \leq \|\square\hat{h}_{k+1-\delta}\|_\infty = \frac{\bar{d}}{2}.$$

Therefore, (20) holds. ∎

Our simulations show that $\|\Delta\hat{b}_k\|_\infty$ and $\|\Delta\hat{h}_k\|_\infty$ are very small; see Fig. 1 in Section V-A. The parametric similarity in the sequential $\mathcal{P}_k$ enables efficient reuse of information from prior steps, which significantly improves the computational efficiency (see Section IV-B).

IV. Fast and Accurate Algorithm

In this section, we propose a fast and accurate algorithm for solving Problem 2, inspired by structural invariance in Section III-A and parametric similarity in Section III-B.

[8]For $\mathcal{Z}_k = Z(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k)$ in $\mathcal{P}_k$, parameters $\hat{G}_k$ and $\hat{A}_k$ are time-invariant, and $\hat{c}_k$ has no effect on the optimal solution.

A. Structural Information-Based Algorithm

Based on the structural invariance of the linear CZ-SMF (Proposition 1), this section proposes a coordinate descent augmented Lagrangian method (CD-AL) for solving $\mathcal{P}_k$. The method precomputes common/invariant parts offline such that the designed online algorithm requires only linear operations.

Recall that $\mathcal{P}_k$ in Problem 2 includes $2n$ LPs with the same constraints but different objectives. We label the LPs with the objectives $\underline{\rho}^{(i)}_k = \min_{\eta,\zeta}(\hat{G}^{(i)}_k\eta + \hat{c}^{(i)}_k)$ and $\bar{\rho}^{(i)}_k = \max_{\eta,\zeta}(\hat{G}^{(i)}_k\eta + \hat{c}^{(i)}_k)$ by $\underline{\mathcal{P}}^{(i)}_k$ and $\overline{\mathcal{P}}^{(i)}_k$, respectively, where $i \in \{1,\ldots,n\}$. Without loss of generality, we only consider $\underline{\mathcal{P}}^{(i)}_k$, and define the shorthand $N = \hat{G}^{(i)}_k$, $M = \hat{c}^{(i)}_k$, $Q = \hat{A}_k$, $p = \hat{b}_k$, and $h = \hat{h}_k$ for the parameters of $\underline{\mathcal{P}}^{(i)}_k$.[9]

Within the augmented Lagrangian framework [20], the augmented Lagrangian function is

$$\begin{aligned}\mathcal{L}_\rho(\eta,\zeta,\alpha,\beta) = N\eta - \alpha(Q\eta - p) - \beta(\eta - \zeta)\\ + \frac{\rho}{2}(\|Q\eta - p\|_2^2 + \|\eta - \zeta\|_2^2),\end{aligned} \quad (23)$$

where $-h \leq \zeta \leq h$, and $\rho > 0$ is the vector of Lagrange multipliers associated with the equality constraints.

Following the augmented Lagrangian method, we update η, ζ, α, and β in (23) by the following iteration framework:

$$\eta^{t+1} = \operatorname*{argmin}_{\eta^t}\mathcal{L}_\rho(\eta^t,\zeta^t,\alpha^t,\beta^t), \quad (24)$$

$$\zeta^{t+1} = \operatorname*{argmin}_{-h\leq\zeta^t\leq h}\mathcal{L}_\rho(\eta^{t+1},\zeta^t,\alpha^t,\beta^t), \quad (25)$$

$$\alpha^{t+1} = \alpha^t - \rho^t(Q\eta^{t+1} - p), \quad (26)$$

$$\beta^{t+1} = \beta^t - \rho^t(\eta^{t+1} - \zeta^{t+1}), \quad (27)$$

where $t = 0, 1, \ldots, N_{\mathrm{out}}$, which involves the minimization step of the primal vector η and ζ [in (24) and (25)] and the update step of α and β [in (26) and (27)]. The iteration terminates if either: 1) the primal residual satisfies $\|Q\eta^t - p\|_\infty \leq \epsilon_{\mathrm{pri}}$ and $\|\eta^t - \zeta^t\|_\infty \leq \epsilon_{\mathrm{con}}$, indicating primal feasibility; or 2) the maximum iteration count N_{out} is reached. Following Nesterov acceleration framework [21], we use the dynamic parameter update $\rho^{t+1} = \frac{1+\sqrt{1+4(\rho^t)^2}}{2}$.

In each iteration t, we need to solve two minimization problems, i.e., (24) and (25). For (24), it is an unconstrained quadratic optimization, the minimizer is

$$\eta^{t+1} = (Q^{\mathrm{T}}Q + I)^{-1}\left(Q^{\mathrm{T}}\left(p + \frac{\alpha^t}{\rho^t}\right) + \zeta^t + \frac{\beta^t - N}{\rho^t}\right).$$

Based on structural invariance from Proposition 1, Q is time-invariant for different $\mathcal{P}_k$ ($k \geq \delta$). Thus, when $k \geq \delta$, we only need to precompute $\Xi = (Q^{\mathrm{T}}Q + I)^{-1}$ once offline[10], making the update process for η consist only of linear operations.

Different from (24), (25) contains a box constraint $-h \leq \zeta \leq h$ in (25). We use the coordinate descent method (CD) to update ζ. The CD method optimizes one variable [w.r.t. component $i \in \{1,\ldots,n^k_g\}$] at a time while keeping the others fixed [22]. More specifically, we employ a randomized cyclic

[9]For $\overline{\mathcal{P}}^i_k$, we can simply set $N = -\hat{G}^{(i)}_k$ and $M = -\hat{c}^{(i)}_k$.

[10]When $k < \delta$, Ξ needs to be recalculated at each k.

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coordinate descent method in [22], where the following n_g^k calculations are performed:

$$\zeta_i^{t+1} = \mathcal{B}_{[-h_i,h_i]}(\eta_i^{t+1} - \frac{1}{\rho^t}\beta_i^t),\ \forall i \in \pi(\{1,\dots,n_g^k\}),$$

where η_i^t, ζ_i^t, β_i^t, and h_i^t denote the ith components of η^t, ζ^t, β^t, and h^t, respectively; $\pi(\cdot)$ represents a random arrangement of coordinates; and $\mathcal{B}_{[-h_i,h_i]}(\cdot)$ is the box-constrained projection operator: $\mathcal{B}_{[-h_i,h_i]}(x) = \max\{-h_i, \min\{x, h_i\}\}$.

The detailed CD-AL method for solving each $\underline{\mathcal{P}}_k^i$ ($\overline{\mathcal{P}}_k^i$) in $\mathcal{P}_k$ (see Problem 2) is provided in Algorithm 1.

Algorithm 1 Structural Information-Based Algorithm

1: **Input:** set $t = 0$, $\eta^0, \zeta^0 \in [-h, h]$, α^0, β^0, and $\rho^0 > 0$; select parameter N_{out}, ϵ_{pri}, ϵ_{con}; offline precomputation $\Xi = (Q^{\mathrm{T}}Q + I)^{-1}$.
2: **while** ($\|Q\eta^t - p\|_\infty > \epsilon_{\text{pri}}$ or $\|\eta^t - \zeta^t\|_\infty > \epsilon_{\text{con}}$) and $t \le N_{\text{out}}$ **do**
3: $\quad \eta^{t+1} = \Xi\left(Q^{\mathrm{T}}\left(p + \frac{\alpha^t}{\rho^t}\right) + \zeta^t + \frac{\beta^t - N}{\rho^t}\right)$;
4: $\quad \zeta_i^{t+1} = \mathcal{B}_{[-h_i,h_i]}(\eta_i^{t+1} - \frac{1}{\rho^t}\beta_i^t),\ \forall i \in \pi(\{1,\dots,n_g^k\})$;
5: $\quad \alpha^{t+1} = \alpha^t - \rho^t(Q\eta^{t+1} - p)$;
6: $\quad \beta^{t+1} = \beta^t - \rho^t(\eta^{t+1} - \zeta^{t+1})$;
7: $\quad \rho^{t+1} \leftarrow \frac{1+\sqrt{1+4(\rho^t)^2}}{2}$;
8: $\quad t = t + 1$;
9: **end while**
10: **Ouput:** Optimal solution $z^* = (\eta^*, \zeta^*)$ and optimal value $N\eta^* + M$.

Leveraging the structural invariance described in Section III-A, Algorithm 1 only involves low-complexity linear operations, thus ensuring high computational efficiency.

B. Warm-Start Strategy Based on Parametric Similarity

For $k \ge \delta$, a warm-start strategy for Algorithm 1 is proposed in this subsection. The strategy generates a warm-start point utilizing the parametric similarity (see Theorem 2) between $\mathcal{P}_k$ and $\mathcal{P}_{k+1}$ to improve the efficiency of solving $\mathcal{P}_{k+1}$.

Let the quadruple $D := (Q, p, N, h)$ be the parameters of $\underline{\mathcal{P}}_k^{(i)}$. We use $\Delta D = (\Delta Q, \Delta p, \Delta N, \Delta h)$ to denote the parametric perturbation between $\underline{\mathcal{P}}_k^{(i)}$ and $\underline{\mathcal{P}}_{k+1}^{(i)}$ such that $D + \Delta D$ is the parameters of $\underline{\mathcal{P}}_{k+1}^{(i)}$. The warm-start strategy generates a warm-start point $z^w = (\eta^w, \zeta^w)$ for $\underline{\mathcal{P}}_{k+1}^{(i)}$ from the optimal solution $z^* = (\eta^*, \zeta^*)$ of $\underline{\mathcal{P}}_k^{(i)}$.[11] Specifically, we compute $\Delta z := (\Delta\eta, \Delta\zeta) = z^w - z^*$ by utilizing the information of ΔD and z^*. Thus, the warm-start point $z^w = z^* + \Delta z$ has fewer iterations than a randomly generated start point for Lines 2-9 in Algorithm 1.

First, $\Delta\eta$ is obtained by solving the optimization problem:

$$\min_{\Delta\eta} \|\Sigma\Delta\eta\|_2^2,\ \text{s.t. } (Q + \Delta Q)(\eta^* + \Delta\eta) = p + \Delta p, \quad (28)$$

where Σ is a positive diagonal matrix, which can be I, $\operatorname{diag}(\eta^*)^{-1}$, or $\operatorname{diag}(\eta^*)^{-2}$ [23]. Due to the structral invariance, $\Delta Q = 0$, and the closed-form solution for (28) is

$$\Delta\eta = (\Sigma^T\Sigma)^{-1}Q^T\left(Q(\Sigma^T\Sigma)^{-1}Q^T\right)^{-1}\Delta p, \quad (29)$$

which is computationally efficient. Note that the constraints of (28) ensure $\|(Q+\Delta Q)\eta^w - (p+\Delta p)\|_\infty = 0 \le \epsilon_{\text{pri}}$. Thus, the condition $\|Q\eta^t - p\|_\infty > \epsilon_{\text{pri}}$ in Line 2 of Algorithm 1 is violated, which can reduce the number of iterations.

[11] In Line 1 of Algorithm 1, (η^0, ζ^0) is replaced by (η^w, ζ^w).

Then, we compute $\Delta\zeta$. To satisfy the box constraint $-h \le \zeta \le h$, we set

$$\Delta\zeta = \mathcal{B}_{[-(h+\Delta h),(h+\Delta h)]}(\eta^* + \Delta\eta) - \zeta^*. \quad (30)$$

The parametric similarity in Theorem 2 implies that $\Delta\eta$ in (29) and $\Delta\zeta$ in (30) are uniformly bounded w.r.t. $k \ge \delta$, which guarantees the efficiency. The detailed strategy is provided in Algorithm 2.

Algorithm 2 Parametric Similarity-Based Warm-Start Strategy

1: **Input:** Optimal point $z^* = (\eta^*, \zeta^*)$, data quadruple $D = (Q, p, N, h)$, perturbation $\Delta D = (\Delta Q, \Delta p, \Delta N, \Delta h)$, selection Σ.
2: $\Delta\eta = (\Sigma^T\Sigma)^{-1}Q^T\left(Q(\Sigma^T\Sigma)^{-1}Q^T\right)^{-1}\Delta p$;
3: $\Delta\zeta = \mathcal{B}_{[-(h+\Delta h),(h+\Delta h)]}(\eta^* + \Delta\eta) - \zeta^*$;
4: **Ouput:** A warm-start point $z^w = z^* + \Delta z$.

Leveraging the parameter similarity from Section III-B, the warm-start point provided by Algorithm 2 accelerates the convergence speed of Algorithm 1.

C. Structural-Parametric Information-Based SMF Algorithm

The detailed design of the fast and accurate linear CZ-SMF is given in Algorithm 3, and we call it the structural-parametric information-based SMF (SPIN-SMF).

If $k < \delta$, we use Algorithm 1 with cold-start strategy to solve $\mathcal{P}_k$ in Problem 2. The cold-start point $z^c = (\eta^c, \zeta^c)$ is constructed as follows:

$$\begin{aligned} \eta^c &= Q^T(Q^TQ)^{-1}p + \sigma_\eta 1, \\ \zeta^c &= \mathcal{B}_{[-h,h]}([Q^T(Q^TQ)^{-1}p + \sigma_\zeta 1]), \end{aligned} \quad (31)$$

where σ_η and σ_ζ are hyperparameters.

If $k \ge \delta$, we couple Algorithm 1 and Algorithm 2 to solve $\mathcal{P}_k$ in Problem 2 efficiently. We present the time and space complexity analysis of the SPIN-SMF algorithm and compare it with the state-of-the-art benchmark algorithms (such as Scott, PCA, OIT-CZ SMF, and Optimal SMF); see Table I.

Algorithm 3 Structural-Parametric Information-Based SMF

Initialization: Bounded $[\![\mathbf{x}_0]\!] = Z(\hat{G}_0^-, \hat{c}_0^-, \hat{A}_0^-, \hat{b}_0^-, \hat{h}_0^-)$ and $k = 0$, select $\delta \in \mathbb{N}_0$.
if $k < \delta$ **then**
  $\mathcal{Z}_k = R_{k,k-1}(\mathcal{Z}_{k-1}) = Z(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k)$;
  $\underline{z}_k^{*,(i)} = (\underline{\eta}_k^{*,(i)}, \underline{\zeta}_k^{*,(i)})$ and $\overline{z}_k^{*,(i)} = (\overline{\eta}_k^{*,(i)}, \overline{\zeta}_k^{*,(i)})$ are solved by Algorithm 1 with cold-start point in (31);
else if $k \ge \delta$ **then**
  $\mathcal{Z}_k = Z(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k)$ with (13) and (16);
  if $k = \delta$ **then**
    $\underline{z}_k^{*,(i)} = (\underline{\eta}_k^{*,(i)}, \underline{\zeta}_k^{*,(i)})$ and $\overline{z}_k^{*,(i)} = (\overline{\eta}_k^{*,(i)}, \overline{\zeta}_k^{*,(i)})$ are solved by Algorithm 1 with cold-start point in (31);
  else if $k > \delta$ **then**
    $\Delta\underline{z}^{(i)}$ and $\Delta\overline{z}^{(i)}$ are solved by Algorithm 2 with $\underline{z}_{k-1}^{*,(i)}$ and $\overline{z}_{k-1}^{*,(i)}$;
    $\underline{z}_k^{*,(i)} = (\underline{\eta}_k^{*,(i)}, \underline{\zeta}_k^{*,(i)})$ and $\overline{z}_k^{*,(i)} = (\overline{\eta}_k^{*,(i)}, \overline{\zeta}_k^{*,(i)})$ are solved by Algorithm 1 with warm-start points $\underline{z}^{w,(i)} = z_{k-1}^{*,(i)} + \Delta\underline{z}^{(i)}$ and $z^{w,(i)} = z_{k-1}^{*,(i)} + \Delta\overline{z}^{(i)}$;
  end if
end if
$\underline{\rho}_k^{(i)} = \hat{G}_k^{(i)}\underline{\eta}_k^{*,(i)} + \hat{c}_k^{(i)}$ and $\overline{\rho}_k^{(i)} = \hat{G}_k^{(i)}\overline{\eta}_k^{*,(i)} + \hat{c}_k^{(i)},\ i = 1,\dots,n$;
$\underline{\rho}_k = [\underline{\rho}_k^1, \dots, \underline{\rho}_k^n]^{\mathrm{T}}$ and $\overline{\rho}_k = [\overline{\rho}_k^1, \dots, \overline{\rho}_k^n]^{\mathrm{T}}$;
$\overline{\mathrm{IH}}(\mathcal{Z}_k) = [\underline{\rho}_k, \overline{\rho}_k] = Z(I_n, \square\hat{c}_k, \square\hat{h}_k)$, where $\square\hat{c}_k = (\underline{\rho}_k + \overline{\rho}_k)/2$ and $\square\hat{h}_k = (\overline{\rho}_k - \underline{\rho}_k)/2$.

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TABLE I
TIME AND SPACE COMPLEXITY

Methods	Time Complexity	Space Complexity
SPIN-SMF	$O(\text{nnz}(Q)+n_g)$	$O((n_g)^2+n_g n_c)$
OR (Scott) [1]	$O((n_g^o+n_c^o)^3+(N)^3)$	$O(nn_g^o+n_c^o n_g^o+(N)^2)$
OR (PCA) [10]	$O(n^2 n_g^o+n^3+(N)^3)$	$O(nn_g^o+n^2+(N)^2)$
OIT-CZ [11]	$O(n^3+(n_g+n_c)^3)$	$O((n+n_c)n_g+(n_g+n_c)^2)$
Optimal SMF	$O(F(k)^3)$	$O(F(k)^2)$

Note: OR denotes the order reduction method. The number of nonzero elements in matrix Q is represented by $\text{nnz}(Q)$. The dimensions are defined as follows: $n_g = n+\delta(p+m)$, $n_c = \delta(p+2m)$, $N = n_g^{\text{tgt}} + n_c^{\text{tgt}}$, and $F(k) = n+k(2p+3m)$ increases with time k. For Scott and PCA, n_g^o and n_c^o denote the generator and constraint dimensions before reduction, while n_g^{tgt} and n_c^{tgt} denote the corresponding target dimensions after reduction.

V. NUMERICAL EXAMPLES

We compare our proposed SPIN-SMF and the state-of-the-art SMFs in this section. The simulations are conducted on a computer with an Intel Core i9-13900KF CPU (3.00 GHz) using MATLAB (.m files).

A. Vehicle Model

We validate the theoretical results for both classical and the proposed SPIN-SMF using the established vehicle benchmark system ($\mathbf{V}_1$) from the comparison study in [4]. The state vector comprises the sideslip angle, yaw rate, yaw angle, lateral position, steering angle, and steering rate. The discretized system ($n=6$, $p=6$, $m=4$) is described by

$$A = \begin{bmatrix} 0.9947 & -0.0079 & 0 & 0 & 0.0210 & 0.0001 \\ 0.3148 & 0.9313 & 0 & 0 & 0.2273 & 0.0007 \\ 0.0015 & 0.0097 & 1 & 0 & 0.0012 & 0 \\ 0.1726 & 0.1442 & 0 & 1 & 0.0197 & 0 \\ 0.2649 & 0.0143 & 0 & 0 & 0.7400 & 0.0043 \\ 39.0353 & 1.9603 & 0 & 0 & -37.7608 & 0.0506 \end{bmatrix}$$

$$B = \text{diag}(0.005,\ 0.055,\ 0.005,\ 0.04,\ 0.02,\ 0.2),$$

$$C = \begin{bmatrix} 0 & 15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix},$$

$$[\![\mathbf{w}_k]\!] = [-1,1]^6,\ [\![\mathbf{v}_k]\!] = [-0.1,0.1]^6,\ [\![\mathbf{x}_0]\!] = \{0_{6\times 1}\}.$$

The measurement matrix C is a linearized measurement model: yaw rate is inferred from lateral acceleration at $15m/s$, and the other outputs are yaw angle, lateral position, and steering angle. The simulations are conducted for 100 runs over the time window $[0, 300]$. We compare the estimation diameter and the computation time of the proposed SPIN-SMF and four classical CZ-SMFs, i.e., the Scott order reduction method [1], the PCA order reduction method [10] (the Scott and PCA methods are given in CORA 2025 [24]), the OIT-CZ SMF method in [11], and the optimal SMF in [13]; see Fig. 1, where the diameter is in the sense of ∞-norm, i.e., $d(\overline{\text{IH}}(\mathcal{Z}_k))$, which can be easily calculated based on the derived $\overline{\text{IH}}(\mathcal{Z}_k)$.

From Fig. 1, we can see that the optimal SMF, the Scott method, and the SPIN-SMF exhibit nearly identical estimation accuracy. However, the SPIN-SMF reduces the computational time by 2 to 4 orders of magnitude compared to the optimal

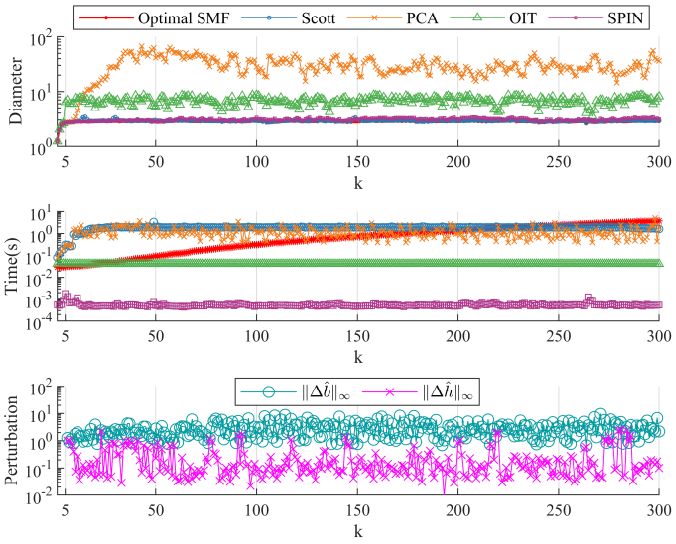


Fig. 1. The diameters of the estimates, computation times, and the perturbations (w.r.t. Theorem 2). The SPIN-SMF runs with $\Sigma = I$, $\rho = 1$, $\delta = n - \text{rank}(C) + 3$, $N_{\text{out}} = 10^6$, and $\epsilon_{\text{pri}} = \epsilon_{\text{con}} = 10^{-4}$ (suitable for most applications); the Scott and PCA order reduction methods are with $n_g = 2(\delta+1)p$, $n_c = (\delta+1)m$, and the degrees-of-freedom order [1] $o_d = (n_g - n_c)/n$; the OIT-CZ SMF method runs with $\bar{\delta} = 5$.

SMF and Scott method. The SPIN-SMF method achieves an average computation time below 1ms while maintaining near-optimal accuracy, demonstrating a significant improvement over the four classical SMF methods. Next, we evaluate the effect of δ on the SPIN-SMF. A larger δ exploits more historical information, producing tighter set estimates (smaller diameter) at the expense of increased computational load. As shown in Table II, increasing δ leads to a gradual decrease in diameter while the runtime increases slightly.

TABLE II
AVERAGED TIME AND DIAMETER OF CZ-SMFS

SMFs	parameter setting	time (ms)	diameter
SPIN	$\delta = 5$	**0.51**	3.0418
	$\delta = 8$	0.89	2.9475
	$\delta = 10$	1.15	2.8944
	$\delta = 20$	2.91	**2.8362**
Scott	$n_g = 72, n_c = 24$	1852.23	2.8794
	$n_g = 108, n_c = 48$	2644.91	2.8517
PCA	$n_g = 72, n_c = 24$	1198.14	30.3184
	$n_g = 108, n_c = 48$	2132.50	22.4688
OIT-CZ SMF	$\bar{\delta} = 5$	41.47	6.7533
	$\bar{\delta} = 20$	62.19	3.0856
Optimal SMF	-	1126.64	2.8155

Note: Scott, PCA, Optimal SMF: any system; OIT-CZ SMF: LTI systems.

To illustrate the efficiency of the SPIN-SMF, Table II also lists other SMFs with different parameters. We can calculate from Table II that the SPIN-SMF reduces computation time by 98.76% ($\delta = 5$) compared to its fastest competitor (OIT-CZ SMF). Simultaneously, the estimation accuracy of the SPIN-SMF method is essentially the same as that of the highest-

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accuracy method (optimal SMF). This shows that the SPIN-SMF proposed in this paper is a fast and accurate method.

B. Monte Carlo Experiment

We conduct Monte Carlo simulations to evaluate the performance of the SPIN-SMF in high-dimensional systems. The system matrices (A, B, C) are randomly generated in MATLAB using `drss` function, ensuring (A, C) is observable. The process and measurement noises are with $[\![\mathbf{w}_k]\!] = [-1, 1]^p$ and $[\![\mathbf{v}_k]\!] = [-1, 1]^m$. The true initial state set is $[\![\mathbf{x}_0]\!] = [-10, 10]^n$, while the estimated initial condition incorporates an offset: $[\![\mathbf{x}_0]\!] = [-10, 10]^n \oplus \{c\}$, where c is randomly generated in $[-1, 1]^n$. Simulation parameters are $n = 10$, $p = m \in \{5, \ldots, 10\}$, and $\delta = n - \mathrm{rank}(C) + 3$. For each $p = m$, 100 trials are conducted. We compare the proposed SPIN-SMF, the SPIN-SMF without warm-start (adopted as an ablation experiment), the Scott method, the PCA method, the OIT-CZ SMF, and optimal SMF under the identical parameter settings as those in Fig. 1. The results are shown in Table III.

TABLE III
TIME AND DIAMETER (AVERAGE) FOR MONTE CARLO EXPERIMENT

SMFs	average time (τ)	average diameter
SPIN	**0.82**ms	**5.1366**
SPIN (No Warm-Start)	3.13ms	5.1369
Scott	5243.51ms	6.5122
PCA	4908.14ms	38.3014
OIT-CZ SMF	64.50ms	6.5276
Optimal SMF	3754.21ms	4.9581

Table III shows that the SPIN-SMF achieves remarkable computational efficiency with high estimation accuracy retained. Specifically, its average computation time reaches the sub-millisecond level ($\tau_{\text{SPIN}} < 1$ms), yielding a 99.98% reduction ($\tau_{\text{SPIN}} = 1.6 \times 10^{-4}\tau_{\text{Scott}}$) against Scott, 99.98% reduction ($1.7\times10^{-4}\tau_{\text{PCA}}$) against PCA, and 98.72% reduction ($1.3\times10^{-2}\tau_{\text{OIT}}$) against OIT-CZ SMF. Ablation results further verify the warm-start's efficacy: a 281.71% time increase is observed without warm start, confirming its convergence acceleration effect. Moreover, SPIN-SMF's estimation diameter is comparable to that of optimal SMF, achieving near-optimal accuracy with far lower computational overhead.

VI. CONCLUSION

In this article, we have proposed a fast and accurate linear CZ-SMF – the SPIN-SMF, inspired by our established two fundamental properties: structural invariance and parametric similarity. For the structural invariance, it provides invariant parameters in constrained zonotopic estimates over time; for the parametric similarity, it guarantees the uniform boundedness of the other parameters (which are time-varying) in consecutive time steps. By leveraging these properties, the SPIN-SMF achieves significant computational efficiency improvements. Specifically, structural invariance enables precomputation of matrix inversions such that our algorithm only involves linear operations; Parametric similarity supports an efficient warm-start strategy by reusing information from the previous time step to initialize our algorithm. For future work, we will explore extending the SPIN-SMF to linear time-varying systems and nonlinear systems.

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